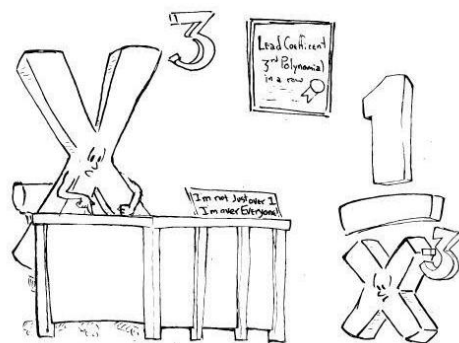


Unit 4: Exponential Functions

Expectations

- evaluate powers with rational exponents, simplify expressions containing exponents, and describe properties of exponential functions represented in a variety of ways;
- make connections between the numeric, graphical, and algebraic representations of exponential functions;
- identify and represent exponential functions, and solve problems involving exponential functions, including problems arising from real-world applications.



Homework Log

Lesson	Date	Section	Assigned Work	Extra Practice
1		4.2 Working with Integer Exponents Part 1	Text: p. 221 #1 – 9(ace), 10, 12, 13, 15, 17, 18	Text: p. 221 #1 – 9 (bdf for each)
2		4.2 Working with Integer Exponents Part 2	LP: p. 5 #1 – 7	Worksheet Online
3		4.3 Working with Rational Exponents	Text: p. 229 #1 – 6, 8, 10 – 14, 18	Worksheet Online
4		4.4 Simplifying Algebraic Expressions Involving Exponents	Text: p. 235 #1, 2, 4 – 6, 9 – 11, 14, 15	Text: p. 236 #3, 7, 8, 12
5		4.5A Exploring the Properties of Exponential Functions	LP: pp. 11 – 12	Text: p. 243 #1, 2
6		4.5B Exploring the Properties of Exponential Functions	LP: pp. 13 – 14	
7		4.6 Transformations of Exponential Functions Part 1	LP: p. 19 – 20	
8		4.6 Transformations of Exponential Functions Part 2	Text: p. 251 #1 – 6, 11 – 14	
9		Determining the Equation of Exponential Functions	LP: p. 23 – 24	Text: p. 252 #9
10		Solving Exponential Equations	LP: p. 26 #1 – 11 (eoo)	LP: p.25 the rest
11		4.7 Applications Involving Exponential Functions	Text: p. 261 #3, 4, 7, 8, 10 LP: p. 31	Text: p. 261 #1, 2, 5, 6, 9, 11 – 16,
12		Review for Test	LP: p. 32 – 34	Text: p. 267 #1, 5, 6, 8, 11, 12, 14, 16, 17

4.2 Reviewing the Exponent Laws

Recall Terminology

a^m is a power in **Exponential Form** where:

m is the exponent

a is the base

a^m is a power of base a

In **Expanded Form**, $a^m = a \cdot a \cdot a \cdot a \cdot \dots$

To **simplify** an expression means to leave the final answer in exponential form.

Recall the Laws

Rule	Algebraic Expression of Rule	Explanation in Words	Example
Multiplication of Powers	$a^m \cdot a^n = a^{m+n}$		$(2m^2n^3)(-3m^4n^6)$
Division of Powers	$a^m \div a^n = a^{m-n}$		$27x^9 \div 3x^{-6}$
Power of a Power	$(a^m)^n = a^{mn}$		$(x^{-5})^{-1}$
Power of a Product	$(a^m b^n)^p = a^{mp} b^{np}$		$(4ab^3)^2$
Power of a Quotient	$\left(\frac{a^m}{b^n}\right)^p = \frac{a^{mp}}{b^{np}}$		$\left(\frac{3x^2}{2y^3}\right)^4$

Example 1 Use the laws of exponents to simplify the following. (Remember that more than one law can be used to simplify an expression completely.)

a) $(4ab^4)(-5a^3b^2)$

b) $(12b^2)(8b^{-4}), (6b^{-10})$

c) $\left(\frac{-1}{2}c^2d^3\right)^4$

d) $\frac{(t^7)^3 \cdot t}{t^{16}}$

Working with Integer Exponents

Negative Exponent	$\left(\frac{a}{b}\right)^{-m} = \left(\frac{b}{a}\right)^m$		$\left(\frac{3}{5}\right)^{-2}$
Zero Exponent	$a^0 = 1$		$(14a^3b^{-4})^0$

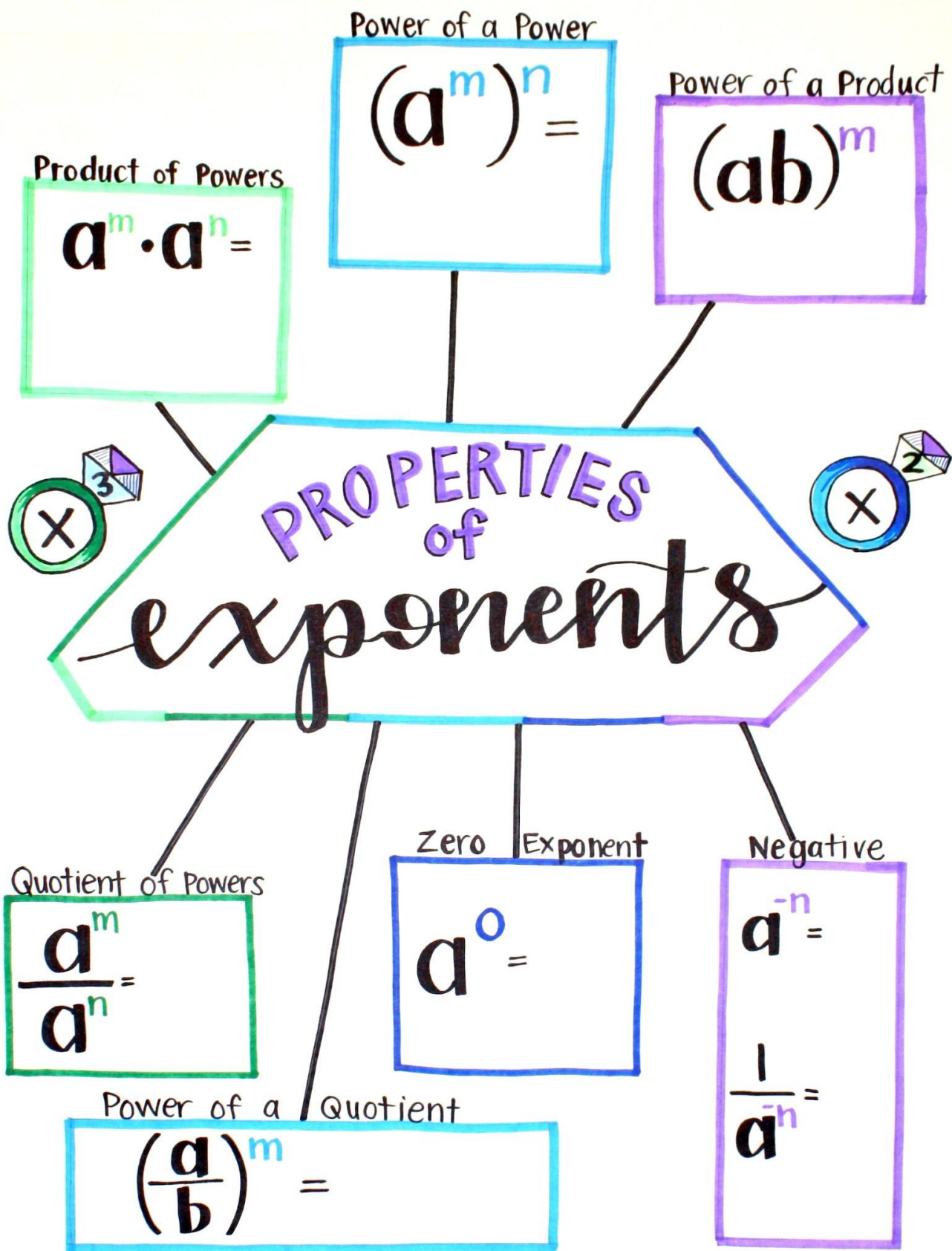
Example 2 Use the laws of exponents to simplify the following.

a) $\frac{(-m^2n^3)^2(mn^{-4})}{(mn^3)^4}$

b) $\frac{x(x^{4a+1})}{x^{a+3}}$

c) $\frac{(3^4 + 2^6)^0}{3^{-1}}$

d) $\frac{2^{-1} + 4^{-2}}{2^{-2} + 4^{-1}}$



Exponent Laws Practice

1. Simplify. Express each answer with positive exponents.

$$\begin{array}{llll} \text{a)} & a^5 \cdot a^6 \cdot a & \text{b)} & (m^2)(m^6) \\ \text{c)} & (n^{-4})(n^{-5}) & \text{d)} & (a^{-3})(b^{-2})(a^2) \\ \text{e)} & 5m^4 \cdot 3m^2 & \text{f)} & 5a(-2ab^2)(-3b^3) \\ \text{g)} & (3x^{-2}y^2)(-2x^2y^{-3}) \end{array}$$

2. Simplify. Express each answer with positive exponents.

$$\begin{array}{lll} \text{a)} & x^6 \cdot x^3 & \text{b)} & t^0 \cdot t^{-5} \\ \text{c)} & y^{-5} \cdot y^{-3} & & \\ \text{d)} & 25k^{15} \cdot 5k^7 & \text{e)} & (-6m^{-4}n^2) \cdot (2m^{-1}n^{-6}) \\ \text{f)} & \frac{(-2x^{-3}y)(-12x^{-4}y^{-2})}{6xy^{-3}} \end{array}$$

3. Simplify. Express each answer with positive exponents.

$$\begin{array}{llll} \text{a)} & (a^2b^3)^4 & \text{b)} & (a^{-1}b^2)^{-2} \\ \text{c)} & \left(\frac{x^2}{y^3}\right)^5 & \text{d)} & \left(\frac{a^{-2}}{b^{-3}}\right)^{-2} \\ \text{e)} & (-3a^3b^{-2})^{-3} & \text{f)} & \left(\frac{6ab^3}{2ab}\right)^3 \\ \text{g)} & \left(\frac{x^3}{y^4}\right)^4 \left(\frac{y^2}{x^4}\right)^3 \end{array}$$

4. Simplify each of the following.

$$\begin{array}{llll} \text{a)} & (x^{p+q})(x^{p-q}) & \text{b)} & (m^{p+q})^{p-q} \\ \text{c)} & \frac{x^{2a}y^{2b}}{x^{a-b}y^{a+b}} & \text{d)} & \frac{(r^x)^4(r^{2x+3y})^3}{(r^{y-2x})^5} \end{array}$$

5. Use brackets to make each statement true. Justify your solution.

$$\text{a)} \quad 2^{-2} \cdot 2^2 + 2^2 - 2^0 = 2^0 \qquad \text{b)} \quad 3^{-4} - 3^{-2} \cdot 3^0 - 3^2 = 3^{-4}$$

6. For which non-zero values of x is $-x^{-4} = (-x)^{-4}$? Explain.

7. For which values of x is $x^{-4} \cdot x^{-4} = 1$ true? Explain.

Solutions:

$$1.\text{a)} \ a^{12} \quad \text{b)} \ m^8 \quad \text{c)} \ \frac{1}{n^9} \quad \text{d)} \ \frac{1}{ab^2} \quad \text{e)} \ 15m^6 \quad \text{f)} \ 30a^2b^5 \quad \text{g)} \ \frac{-6}{y} \quad 2.\text{a)} \ x^3 \quad \text{b)} \ t^5 \quad \text{c)} \ \frac{1}{y^2} \quad \text{d)} \ 5k^8 \quad \text{e)} \ \frac{-3n^8}{m^3} \quad \text{f)} \ \frac{4y^2}{x^8}$$

$$3.\text{a)} \ a^8b^{12} \quad \text{b)} \ \frac{a^2}{b^4} \quad \text{c)} \ \frac{x^{10}}{y^{15}} \quad \text{d)} \ \frac{a^4}{b^6} \quad \text{e)} \ \frac{b^6}{-27a^9} \quad \text{f)} \ 27b^6 \quad \text{g)} \ \frac{1}{y^{10}} \quad 4.\text{a)} \ x^{2p} \quad \text{b)} \ m^{p^2-q^2} \quad \text{c)} \ x^{a+b}y^{-a+b} \quad \text{d)} \ r^{20x+4y}$$

5.a) $2^{-2} \cdot (2^2 + 2^2) - 2^0 = 2^0$ b) $(3^4 - 3^{-2}) \cdot (3^0 - 3^2) = 3^{-4}$ 6. There is no non-zero value of x that will make that statement true, since the left-hand side is negative and the right-hand side of the equation is positive. 7. All value of x will satisfy.

4.3 Working with Rational Exponents

Examples: Evaluate each of the following.

a) $9^{\frac{1}{2}} \cdot 9^{\frac{1}{2}}$

b) $\sqrt{9} \cdot \sqrt{9}$

c) $8^{\frac{1}{3}} \cdot 8^{\frac{1}{3}} \cdot 8^{\frac{1}{3}}$

b) $\sqrt[3]{8} \cdot \sqrt[3]{8} \cdot \sqrt[3]{8}$

Therefore, _____

Therefore, _____

In general: _____

In general: _____

Exponents in the form $\frac{1}{n}$:

Ex. 1: Write each of the following in radical form. Evaluate, if possible.

a) $64^{\frac{1}{2}}$

b) $27^{-\frac{1}{3}}$

c) $(-243)^{\frac{1}{5}}$

d) $16^{\frac{1}{4}}$

e) $(-625)^{\frac{1}{4}}$

Ex. 2: Write each of the following in exponent form. Evaluate, if possible.

a) $\sqrt[3]{-343}$

b) $-\sqrt{121}$

c) $\sqrt[4]{16}$

d) $\sqrt[3]{64}$

e) $\frac{1}{\sqrt[5]{-32}}$

Take Note:

Exponents in the form $\frac{m}{n}$

Use the exponent laws to express $x^{\frac{2}{3}}$ in TWO ways. (Recall: Power Law _____)

i) $x^{\frac{2}{3}}$

ii) $x^{\frac{2}{3}}$

Exponents in the form $\frac{m}{n}$:

Ex. 3: Write each of the following in radical form. Evaluate, if possible.

a) $8^{\frac{2}{3}}$

b) $-25^{\frac{5}{2}}$

c) $256^{-\frac{3}{4}}$

d) $64^{-1.5}$

e) $(-27)^{-\frac{2}{3}}$

Ex. 4: Simplify to a single positive exponent.

a) $\left(2^{\frac{2}{3}}\right)\left(2^{-\frac{1}{2}}\right)$

b) $\frac{3^{\frac{-3}{4}}}{3^2}$

c) $5^{\frac{1}{2}} \times 5^{-\frac{1}{4}} \cdot 5^{\frac{3}{8}}$

d) $\left(2^{\frac{5}{6}}\right)^{-3} \times \left(2^{-3}\right)^{\frac{1}{4}}$

4.4 Simplifying Algebraic expressions involving exponents

Simplify. Express final answers with positive exponents where necessary.

a) $\sqrt{81m^8}$

b) $\left(\frac{-8x^3}{216}\right)^{-\frac{1}{3}}$

c) $\left(\frac{16}{81y^8}\right)^{-\frac{3}{4}}$

d) $\sqrt{\sqrt{x^{12}}}$

e) $\sqrt[3]{\sqrt{y^4}}$

f) $\sqrt{\sqrt{10000y^6}}$

g) $\sqrt[3]{\sqrt{64x^{18}}}$

h) $\sqrt[5]{\frac{\sqrt{x}\sqrt{x^3}}{x^{\frac{3}{4}}}}$

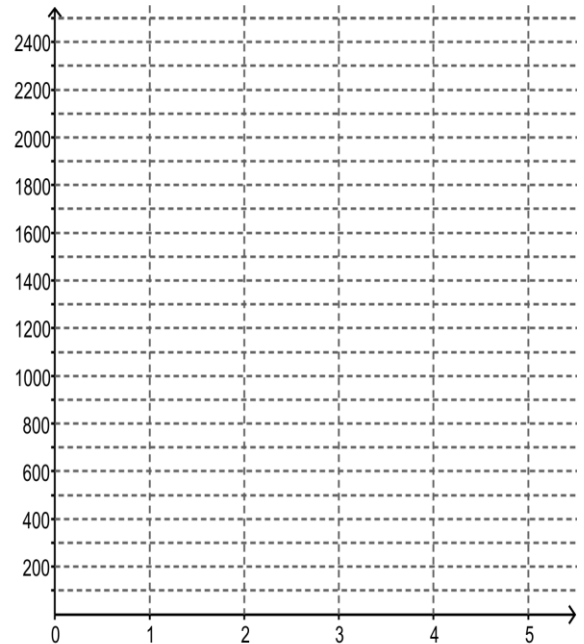
4.5a The Exponential Relation

Recall: A constant first difference indicates a _____ relationship and a constant second difference indicates a _____ relationship.

Exponential relationships have a constant _____.

1. Scientists investigating bacteria growth recorded the number of a certain bacteria every hour.

<i>Time, h</i>	<i>Number of Bacteria</i>	<i>Ratio</i>
0	50	
1	150	
2	450	
3	1350	



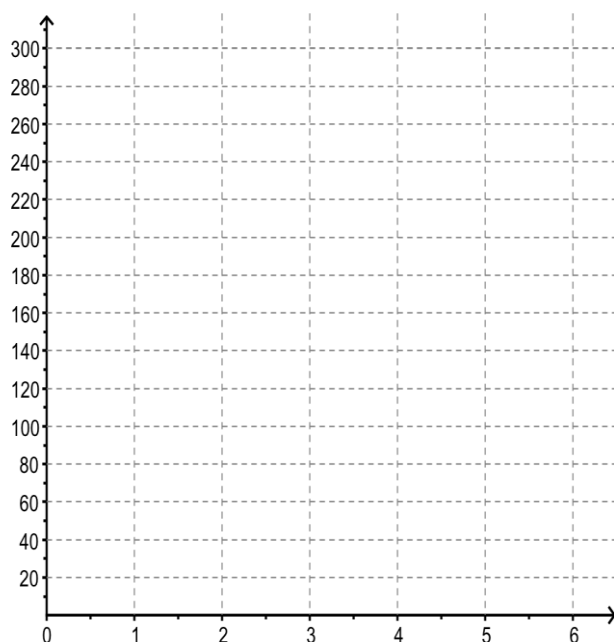
a) How is this scenario an example of exponential growth?

b) How many bacteria will there be after 6 hours?

2. A ball is dropped from a height of 300 cm. After each bounce, the ball's height is recorded in the table below.

a) Graph the ball's height after 5 bounces.

Bounce	Ball's Height (cm)	Ratio
0	300	
1	195	
2	126.75	
3	82.39	
4	53.55	
5	34.81	



b) How is the ball's height an example of exponential decay?

c) What is the height of the ball, to the nearest centimetre, after the 8th bounce?

Investigate Exponential Relationships

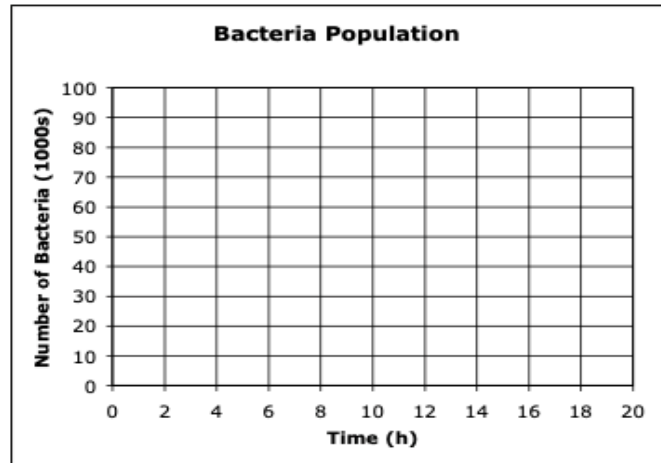
1. For each question:
 - Create a table of values
 - Classify as Linear/Quadratic/Exponential and explain your choice.
 - a) James stacks cans for a grocery store display. The top row has 1 can, the second row has 2 cans, the third row has 3 cans, etc. How many cans are there in the 10th row?
 - b) A soccer ball is kicked. At 1 sec., its height is 20.6m. At 2 sec., its height is 30.4 m. At 3 sec., its height is again at 30.4m. At 4 sec., the height is 20.6m. At 5 sec., the height is 1m.
 - c) A vase that cost \$800 is expected to increase in value by 7% each year for 5 years.
 - d) A \$600 investment is worth \$618 after 1 year, \$636 after 2 years, \$654 after 3 years and \$672 after 4 years.
2. In an old story, *The King's Chessboard*, a king rewards a farmer for saving the kingdom. The king gives the farmer 1 grain of rice on the first day, 2 grains of rice on the second day, 4 on the third day, and so on, doubling the number of grains of rice each day, for 64 days.
 - a) Make a table showing the number of grains of rice the farmer receives each day for the first 7 days.
 - b) How does this story illustrate exponential growth?
 - c) How many grains of rice will the farmer receive on the sixteenth day?
 - d) No matter how rich the king is, he will eventually run out of rice. Explain why?
3. On a television game show, the cash prizes were designed to resemble exponential growth. The prizes are:

\$100	\$2000	\$64 000
\$200	\$4000	\$125 000
\$300	\$8000	\$250 000
\$500	\$16 000	\$500 000
\$1000	\$32 000	\$1 000 000

 - a) Do these cash prizes increase exponentially? Explain your answer.
 - b) Make a new table of 15 cash prizes that do grow exponentially.

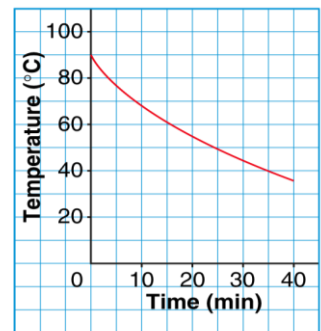
4. The table shows the number of bacteria in 1 cm³ of a bacteria culture over a period of hours.

Time (h)	Number of Bacteria (1000s)
2.5	10.07
2.8	13.07
5.4	14.59
6.5	20.70
9.2	27.94
9.5	31.50
11.0	40.04
13.3	49.90
14.6	64.72
16.4	75.57

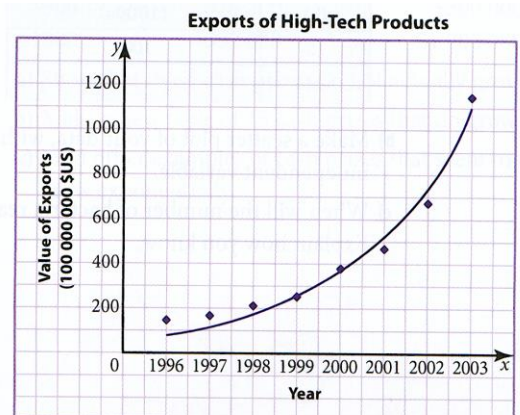


- Make a scatter plot of the table, with time as the independent variable (x axis). Describe the shape of the graph.
 - Estimate the number of bacteria present at the beginning of the test. Explain how you got your answer.
 - Estimate the number of bacteria present after 10 h.
5. A student in a chemistry laboratory heated a liquid chemical to 90°C and let it cool at room temperature to 60°C. This took 15 min. The graph shows the temperature of the chemical (°C) versus time (min).

- Describe the shape of the graph.
- How long did it take for the chemical to cool to 40°C?
- The temperature of the chemical will stabilize when it reaches room temperature (25°C). Estimate how long it will take the liquid to reach room temperature?



6. The graph shows a scatter plot and curve of best fit of the total high-tech exports from China, in hundreds of millions of US dollars, between 1996 and 2003.
- Do you think the graph is an example of exponential growth? Explain.
 - Assume the trend continued. What will be the value of high-tech exports in 2004? 2007?
 - Do you think this trend will continue? Explain.



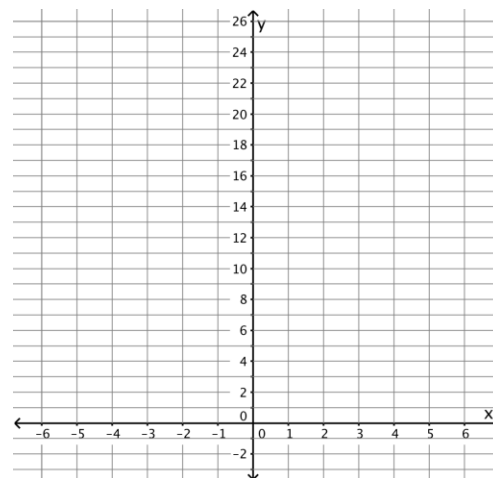
4.5b Investigating the Exponential Function $f(x) = b^x$

Go to: www.desmos.com/calculator

Part A

1. Graph of $y = 2^x$.

- Identify the value of the y-intercept.
Label the y-intercept on your sketch.
- Click on the graph and, move the arrow left.
Watch the y-values as the cursor moves.
What do you notice about these values?



c) As x increases in value, what do you notice about the y-values?

d) What is the domain and range of $y = 2^x$?

2. Graph $y = 3^x$, $y = 5^x$, and $y = 10^x$. Sketch the three functions on the same grid, labelling the functions carefully.

a) What are the common characteristics of these curves?

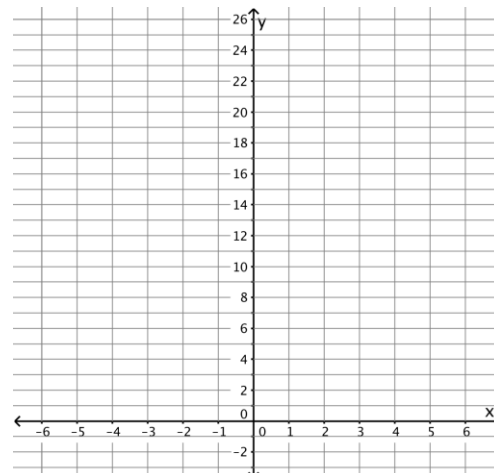
b) As b increases in value, _____.

c) Predict where the graph of the curve $y = 7^x$ will lie relative to the curves graphed earlier. Check your prediction with Desmos.

3. Graph the function $y = \left(\frac{1}{2}\right)^x$.

a) Identify the value of the y-intercept.
Label the y-intercept on your sketch.

b) As x increases in value, what do you notice about the y-values?



c) What is the domain and range of $y = \left(\frac{1}{2}\right)^x$?

4. Graph $y = \left(\frac{1}{3}\right)^x$, $y = \left(\frac{1}{5}\right)^x$, and $y = \left(\frac{1}{10}\right)^x$. Sketch the three functions on the same grid, labelling the functions carefully.

a) What are the common characteristics of these curves?

b) As b decreases in value, _____.

c) Predict where the graph of the curve $y = \left(\frac{1}{7}\right)^x$ will lie relative to the curves graphed earlier. Check your prediction with Desmos.

5. Write a general description of the graph of the exponential curve $y = b^x$.

4.6 Transformations of Exponential Functions

Given $f(x) = b^x$, the transformed function is $g(x) = a(b)^{k(x-d)} + c$.

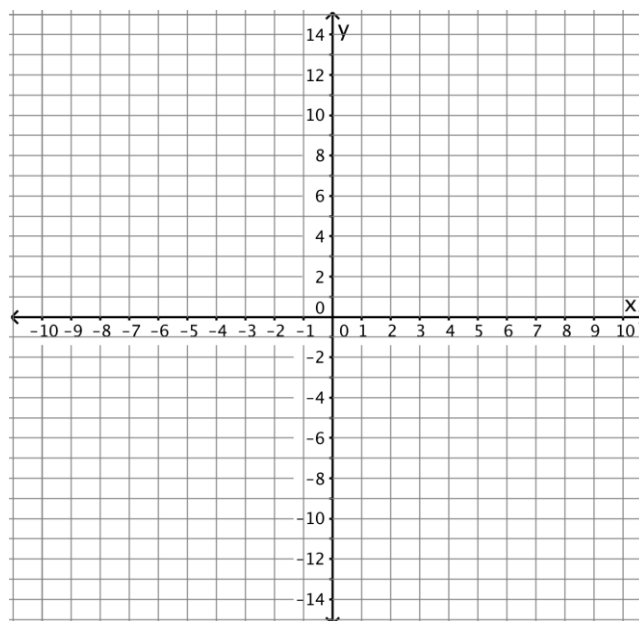
The Role of Parameter

- a - _____
- k - _____
- d - _____
- c - _____

1. a. Use the graph of $f(x) = 2^x$ to sketch the graph of $g(x) = 2^x + 4$ and $h(x) = 2^x - 3$.
Draw a sketch and label each function.

$f(x) = 2^x$		$\Rightarrow$	$g(x) = 2^x + 4$	
x	y		x	y
-2	$\frac{1}{4}$			
-1	$\frac{1}{2}$			
0	1			
1	2			
2	4			

$f(x) = 2^x$		$\Rightarrow$	$h(x) = 2^x - 3$	
x	y		x	y
-2	$\frac{1}{4}$			
-1	$\frac{1}{2}$			
0	1			
1	2			
2	4			



- b. State the domain, range, y-intercept and asymptote of each function.

$g(x) = 2^x + 4$

- Domain: _____
- Range: _____
- Y-intercept: _____
- Asymptote: _____

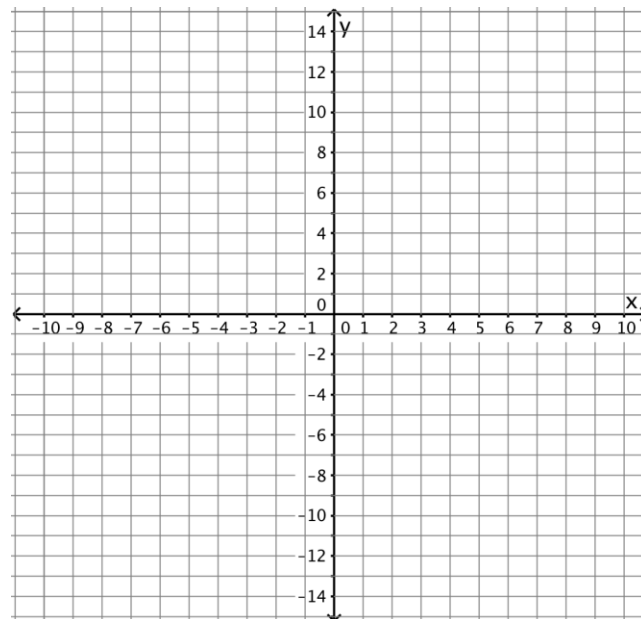
$h(x) = 2^x - 3$

- Domain: _____
- Range: _____
- Y-intercept: _____
- Asymptote: _____

2. a. Use the graph of $f(x) = 3^x$ to sketch the graph of $g(x) = 3^{x-2}$ and $h(x) = 3^{x+3}$.
Draw a sketch and label each function.

$f(x) = 3^x$		$\Rightarrow$	$g(x) = 3^{x-2}$	
x	y		x	y
-2	$1/9$			
-1	$1/3$			
0	1			
1	3			
2	9			

$f(x) = 3^x$		$\Rightarrow$	$h(x) = 3^{x+3}$	
x	y		x	y
-2	$1/9$			
-1	$1/3$			
0	1			
1	3			
2	9			



- b. State the domain, range, y-intercept and asymptote of each function.

$$g(x) = 3^{x-2}$$

- Domain: _____
- Range: _____
- Y-intercept: _____
- Asymptote: _____

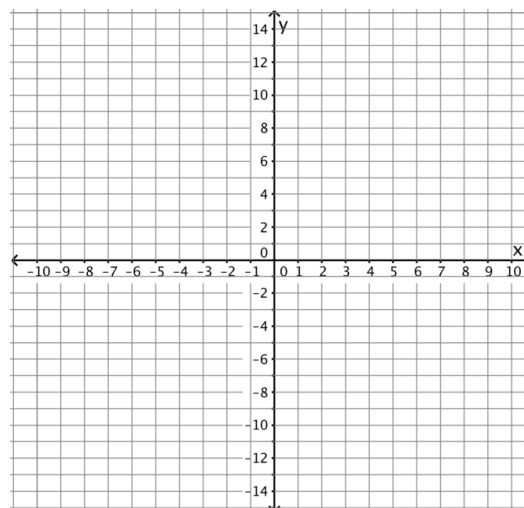
$$h(x) = 3^{x+3}$$

- Domain: _____
- Range: _____
- Y-intercept: _____
- Asymptote: _____

3. a. Use the graph of $f(x) = 5^x$ to sketch the graph of $g(x) = 2(5^x)$ and $h(x) = \frac{1}{2}(5^x)$.
Draw a sketch and label each function.

$f(x) = 5^x$		$\Rightarrow$	$g(x) = 2(5^x)$	
x	y		x	y
-1	$1/5$			
0	1			
1	5			

$f(x) = 5^x$		$\Rightarrow$	$h(x) = 1/2(5^x)$	
x	y		x	y
-1	$1/5$			
0	1			
1	5			



b. State the domain, range, y-intercept and asymptote of each function.

$$g(x) = 2(5^x)$$

- Domain: _____
- Range: _____
- Y-intercept: _____
- Asymptote: _____

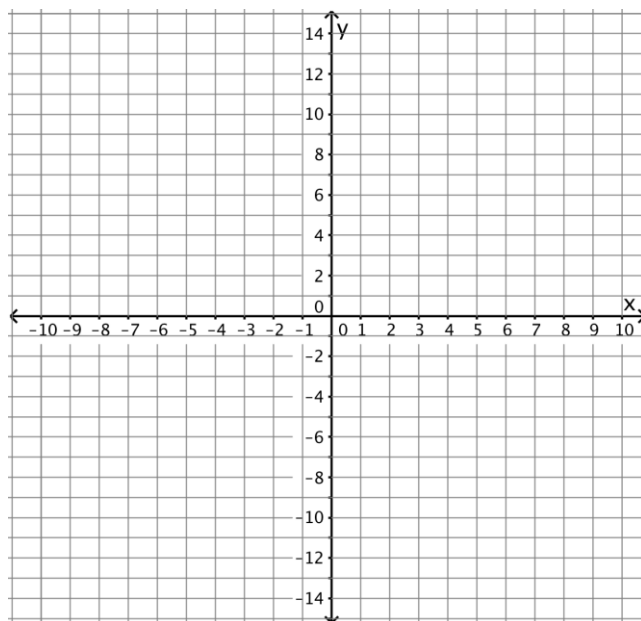
$$h(x) = 1/2(5^x)$$

- Domain: _____
- Range: _____
- Y-intercept: _____
- Asymptote: _____

4. a. Use the graph of $f(x) = 10^x$ to sketch the graph of $g(x) = 10^{2x}$ and $h(x) = 10^{\frac{x}{2}}$.
Draw a sketch and label each function.

$f(x) = 10^x$		$g(x) = 10^{2x}$	
x	y	x	y
-1	1/10		
0	1		
1	10		

$f(x) = 10^x$		$h(x) = 10^{x/2}$	
x	y	x	y
-1	1/10		
0	1		
1	10		



b. State the domain, range, y-intercept and asymptote of each function.

$$g(x) = 10^{2x}$$

- Domain: _____
- Range: _____
- Y-intercept: _____
- Asymptote: _____

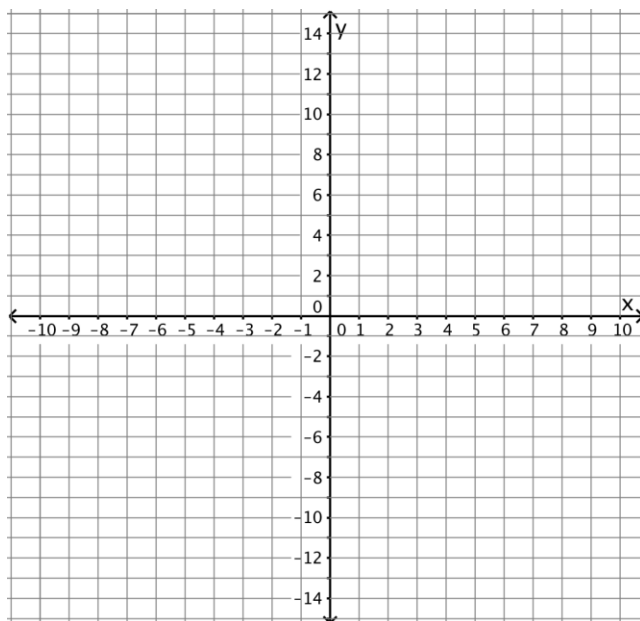
$$h(x) = 10^{x/2}$$

- Domain: _____
- Range: _____
- Y-intercept: _____
- Asymptote: _____

5. a. Use the graph of $f(x) = 2^x$ to sketch the graph of $g(x) = 2^{-x}$ and $h(x) = -2^x$.
Draw a sketch and label each function.

$f(x) = 2^x$			$g(x) = 2^{-x}$	
x	y		x	y
-2	$\frac{1}{4}$			
-1	$\frac{1}{2}$			
0	1			
1	2			
2	4			

$f(x) = 2^x$			$h(x) = -2^x$	
x	y		x	y
-2	$\frac{1}{4}$			
-1	$\frac{1}{2}$			
0	1			
1	2			
2	4			



- b. State the domain, range, y-intercept and asymptote of each function.

$$g(x) = 2^{-x}$$

- Domain: _____
- Range: _____
- Y-intercept: _____
- Asymptote: _____

$$h(x) = -2^x$$

- Domain: _____
- Range: _____
- Y-intercept: _____
- Asymptote: _____

Transformations of Exponential Functions Practice

- Sketch the graph of $f(x) = 2^{x-1} + 3$.
 - State the asymptote, the domain and the range of the function.
- For each of the following, state
 - the equation of the horizontal asymptote,
 - whether the function is increasing or decreasing,
 - the y-intercept
 - $y = 3^x - 5$
 - $y = 2^x + 4$
 - $y = 4\left(\frac{1}{3}\right)^x$
 - $y = \left(\frac{1}{2}\right)^x + 2$
 - $y = 2(5^x) - 1$
 - $y = 5\left(\frac{2}{3}\right)^x + 1$
- On the same axes, sketch each of the following.
 - $y = 2^x$
 - $y = -(2^x)$
 - $y = 2^{-x}$
 - $y = -(2^{-x})$
- Graph each of the following functions. Describe each transformation.
 - $f(x) = 2^{x-1} - 2$
 - $f(x) = 3^{x+2} + 1$
 - $f(x) = 4^{x+1} - 3$
 - $f(x) = 2^{-(x+1)} - 2$
 - $f(x) = 3^{-(x-2)} + 1$
 - $f(x) = 4^{-(x-1)} - 3$
- Graph the following pairs of functions on the same axes. What do you notice?
 - $f(x) = 9^x$ and $g(x) = 3^{2x}$
 - $f(x) = 8^x$ and $g(x) = 2^{3x}$
 - $f(x) = \left(\frac{1}{3}\right)^x$ and $g(x) = 3^{-x}$
 - $f(x) = \left(\frac{1}{4}\right)^x$ and $g(x) = 2^{-2x}$
- For the curve $y = 3(4^x) + 5$,
 - determine
 - the horizontal asymptote,
 - the y-intercept,
 - whether the curve is increasing or decreasing,
 - the domain and range.
 - Sketch the curve.
- For the curve $y = 2\left(\frac{1}{3}\right)^x - 4$,
 - determine
 - the horizontal asymptote,
 - the y-intercept,
 - whether the curve is increasing or decreasing,
 - the domain and range.
 - Sketch the curve.

8. Graph each of the following functions.

a) $f(x) = 3(2^{-x+2}) - 1$

b) $f(x) = -2(3^{-x-1}) + 3$

c) $f(x) = -\left(\frac{1}{2}\right)^{-x+3} - 2$

d) $f(x) = 2\left(\frac{1}{3}\right)^{-x-2} + 1$

e) $f(x) = 2^{\frac{1}{2}x+1} - 2$

f) $f(x) = 3^{\frac{-1}{2}x-2} + 1$

g) $f(x) = \left(\frac{1}{2}\right)^{\frac{1}{2}x-1}$

h) $f(x) = \left(\frac{1}{3}\right)^{-\frac{1}{2}x-2}$

Determining the Equation of an Exponential Function

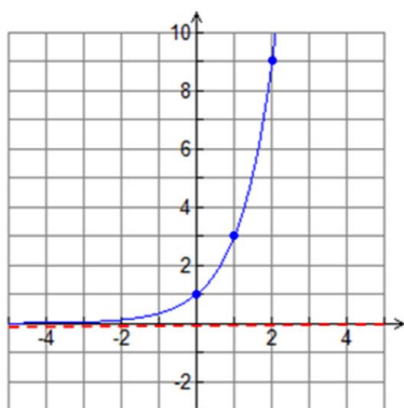
Steps to determining the equation of an exponential function.

1. Identify the asymptote. That will be the c value.
2. Count from the asymptote to the first (closest) point on the graph. That will be the transformed y-intercept point. The x-coordinate of this point is the d value.
3. Count from the asymptote to the second point on the graph. This will be your base (whole number).
4. If the graph is completely below the x-axis, then the graph has a $-a$ value.
5. If the graph is above the asymptote and is increasing, then k is a positive value. If the graph is above the asymptote and is decreasing, then k is a negative value. If the graph is below the asymptote and is increasing, then k is a negative value. If the graph is below the asymptote and is decreasing, then k is a positive value.

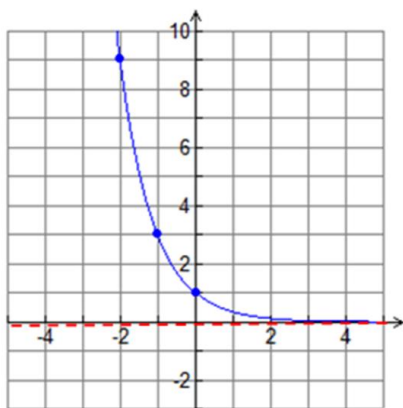
Note: Vertical and horizontal stretches and compressions will not be affecting these graphs.

1. Determine a possible equation for each of the following exponential functions:

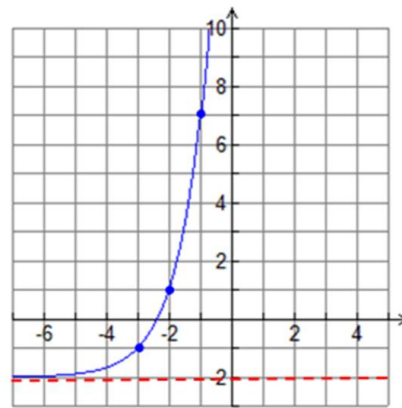
a)



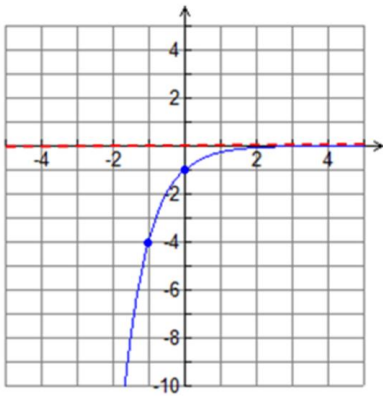
b)



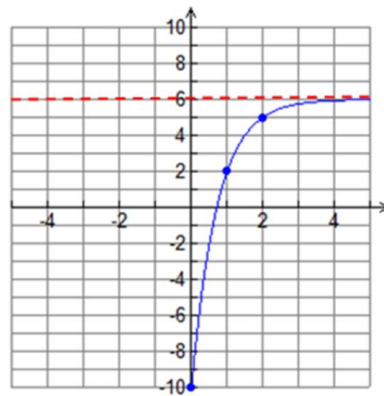
c)



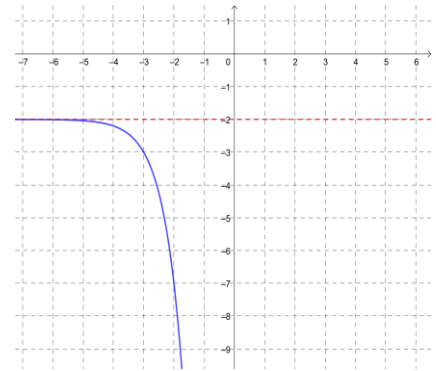
d)



e)



f)



2. The graph of $f(x) = 3^x$ is compressed vertically by a factor of $1/2$, reflected in the y -axis, and translated right 4 units and downward 5 units.

a) Write the equation of the new function.

b) State the domain, range, y -intercept and equation of the horizontal asymptote.

3. The equation of the function that represents $f(x) = \left(\frac{1}{4}\right)^x$ after it is compressed horizontally by a factor of $1/2$, reflected in the x -axis, and shifted 4 to the left and 6 units up.

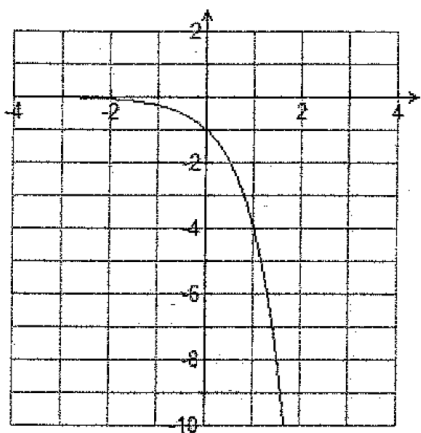
a) Write the equation of the new function.

b) State the domain, range, and equation of the horizontal asymptote.

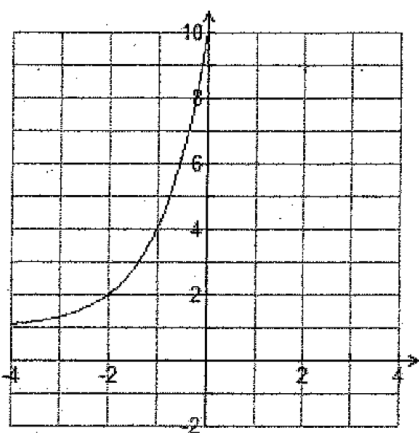
Finding the Equation of the Exponential Function - Practice

1. Determine a possible equation for each of the following exponential functions:

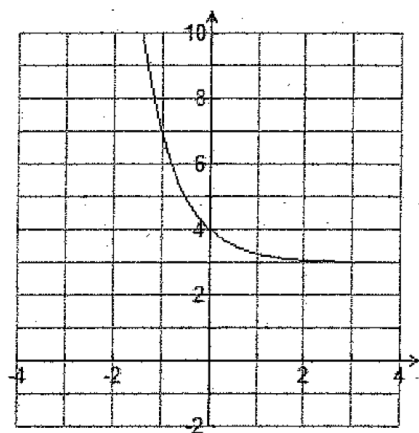
a)



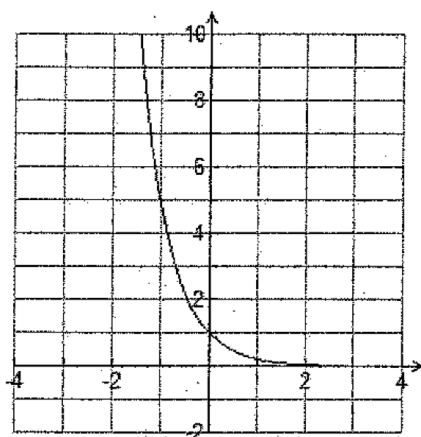
b)



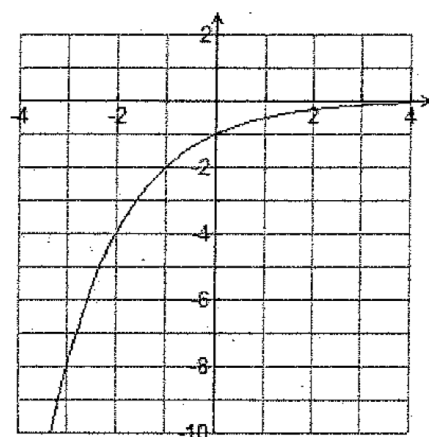
c)



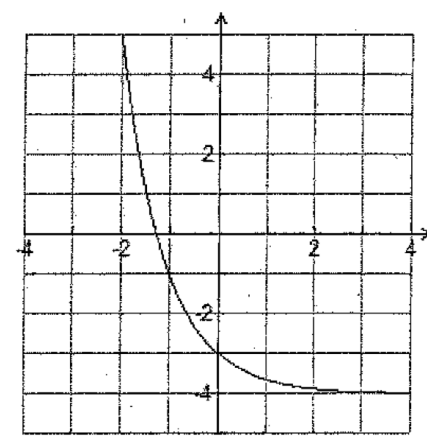
d)



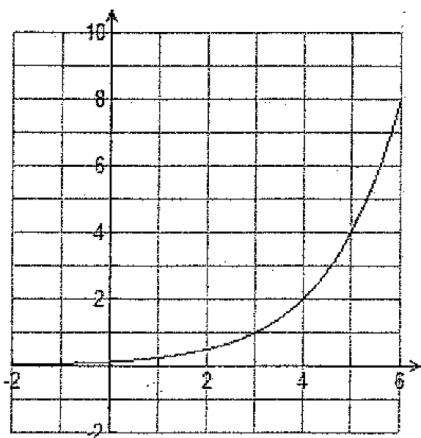
e)



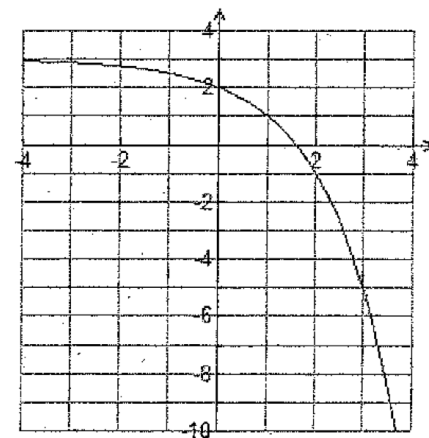
f)



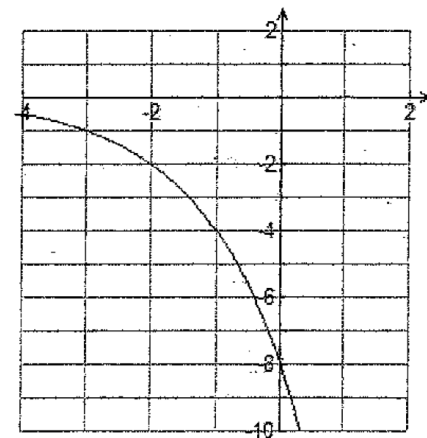
g)



h)



i)



2. State the equation of each of the following. Also, state the domain, range, y-intercept, and equation of the asymptote.
- The graph of $f(x) = 2^x$ is stretched vertically by a factor of 3, reflected in the x-axis, and translated to the right 5 units and up 10 units.
 - The graph of $f(x) = \left(\frac{1}{2}\right)^x$ is stretched horizontally by a factor of 1.5, reflected horizontally, and translated 3 units down, and 4 units left.
 - The graph of $f(x) = 3^x$ is stretched vertically by a factor of 2 and compressed horizontally by a factor of $1/2$. It is also, reflected in the y-axis and translated 2 units down and 4 units right.
3. State the parent functions of the following functions. Also, describe the transformations in the order they would be applied.
- $f(x) = 2(3^x) + 4$
 - $f(x) = -2^{x+1} - 3$
 - $f(x) = 4^{3x+2}$
 - $f(x) = 2\left(\frac{1}{3}\right)^{\frac{1}{2}x+1}$
 - $f(x) = \left(\frac{2}{3}\right)^{-x+3} + 6$

Solving Exponential Equations

An *exponential equation* is one in which the _____ is contained within an exponent or exponents.

As with all types of equations, algebra can be used to determine an exact solution for an exponential equation. When the _____ on either side of the equation have the same base, the exponents can be set equal and the resulting equation solved.

In other words:



Examples: Solve each of the following:

a) $7^x = 49$

b) $\left(\frac{1}{6}\right)^{-x} = 216$

c) $2^{3x+5} = 2^{4x-3}$

d) $5^{2x-1} = \frac{1}{125}$

e) $3^x = 9^{x-1}$

f) $4(2^x) = 32$

g) $9^{2x+1} = 81(27^x)$

h) $\left(\frac{1}{8}\right)^4 = 16^{3x-1}$

i) $2^{x+2} - 2^x = 48$

Solving Exponential Equations

Practise

A

1. Solve.

- a) $2^x = 16$
- c) $2^x = 128$
- e) $4^y = 256$
- g) $(-3)^x = -27$
- i) $(-5)^a = 25$
- k) $-2^x = -16$
- m) $-5^x = -625$
- o) $(-1)^m = -1$

- b) $3^x = 27$
- d) $5^x = 125$
- f) $729 = 9^z$
- h) $(-2)^x = -32$
- j) $81 = (-3)^x$
- l) $-4^y = -64$
- n) $(-1)^x = 1$

2. Solve.

- a) $7^{w-2} = 49$
- c) $2^{1-x} = 128$
- e) $5^{3x-1} = 25$
- g) $4^{x-1} = 1$
- i) $(-1)^{2x} = 1$

- b) $3^{x+4} = 27$
- d) $4^{3k} = 64$
- f) $-81 = -3^{2x+8}$
- h) $3^{2-2x} = 1$

3. Solve and check.

- a) $6^{x+3} = 6^{2x}$
- c) $3^{2y+3} = 3^{y+5}$
- e) $7^{5d-1} = 7^{2d+5}$

- b) $2^{x+3} = 2^{2x-1}$
- d) $2^{4x-7} = 2^{2x+1}$
- f) $3^{b-5} = 3^{2b-3}$

4. Solve.

- a) $16^{2x} = 8^{3x}$
- c) $27^{x-1} = 9^{2x}$
- e) $16^{2p+1} = 8^{3p+1}$

- b) $4^t = 8^{t+1}$
- d) $25^{2-c} = 125^{2c-4}$
- f) $(-8)^{1-2x} = (-32)^{1-x}$

5. Solve and check.

- a) $2^{x+5} = 4^{x+2}$
- c) $9^{2q-6} = 3^{q+6}$
- e) $27^{y-1} = 9^{2y-4}$

- b) $2^x = 4^{x-1}$
- d) $4^x = 8^{x+1}$
- f) $8^{x+3} = 16^{2x+1}$

6. Solve and check.

- a) $5^{4-x} = \frac{1}{5}$
- c) $6^{3x-7} = \frac{1}{6}$
- e) $5^{2n+1} = \frac{1}{125}$

- b) $10^{y-2} = \frac{1}{10\,000}$
- d) $3^{3x-1} = \frac{1}{81}$
- f) $\frac{1}{256} = 2^{2-5w}$

7. Solve and check.

- a) $4^x = 8$
- c) $(-8)^y = -2$
- e) $2^{9x} = \frac{1}{8}$
- g) $2^x = 16^4$
- i) $9^{2s+1} = 27$
- b) $64^z = 16$
- d) $9^{-x} = 3$
- f) $9^{6x} = \frac{1}{27}$
- h) $2^{-2g} = 32$

8. Solve and check.

- a) $9^{x+1} = 27^{2x}$
- c) $36^{t-2} = 216^{-2t}$
- e) $25^{1-3x} = 125^{-x}$
- b) $16^y = 64^{2y-1}$
- d) $8^{2x-1} = 16^{x-1}$
- f) $16^{3+k} = 32^{1-2k}$

9. Solve and check.

- a) $5 = 25^{\frac{x}{2}}$
- c) $9^{\frac{y}{5}} = 27$
- e) $4^{\frac{x}{4}} = \frac{1}{8}$
- b) $8 = 2^{\frac{x}{3}}$
- d) $\frac{1}{2} = 2^{\frac{a}{3}}$
- f) $\left(\frac{3}{2}\right)^{\frac{m}{2}} = \frac{4}{9}$

10. Solve.

- a) $3(5^{x+1}) = 15$
- b) $2(3^{y-2}) = 18$
- c) $5(4^x) = 10$
- d) $2(4^{v+1}) = 1$
- e) $2 = 6(3^{4f-2})$
- f) $27(3^{3x+1}) = 3$

11. Solve and check.

- a) $2^{x+2} - 2^x = 48$
- b) $4^{x+3} + 4^x = 260$
- c) $2^{a+5} + 2^a = 1056$
- d) $6^{x+1} + 6^{x+2} = 7$
- e) $3^{x+3} - 3^{x+1} = 648$
- f) $10^{z+4} + 10^{z+3} = 11$
- g) $2^{x+2} - 2^{x+5} = -7$
- h) $3^{m+1} + 3^{m+2} - 972 = 0$
- i) $5^{n+2} - 5^{n+3} = -2500$

Answers

1. a) 4 b) 3 c) 7 d) 3 e) 4 f) 3 g) 5 h) 2 i) 4 k) 4 l) 3 m) 4
- n) x any even integer o) m any odd integer
2. a) 4 b) -1 c) -6
- d) 1 e) 1 f) -2 g) 1 h) 1 i) all values of x
3. a) 3 b) 4 c) 2 d) 4
- e) 2 f) -2 g) 0 h) -3 i) -3 j) 2 k) 1 l) -2
4. a) 1 b) 2 c) 6
- d) -3 e) 5 f) 1 g) 5 h) -2 i) -2 j) 2 k) 7
5. a) $\frac{3}{2}$ b) $\frac{1}{3}$ c) $\frac{1}{2}$ d) $\frac{1}{3}$ e) $\frac{1}{4}$ f) $\frac{1}{2}$ g) $\frac{1}{4}$ h) $\frac{1}{2}$ i) $\frac{1}{4}$ j) $\frac{1}{2}$ k) $\frac{1}{4}$ l) $\frac{1}{2}$ m) $\frac{1}{4}$ n) $\frac{1}{2}$ o) $\frac{1}{4}$ p) $\frac{1}{2}$ q) $\frac{1}{4}$ r) $\frac{1}{2}$ s) $\frac{1}{4}$ t) $\frac{1}{2}$ u) $\frac{1}{4}$ v) $\frac{1}{2}$ w) $\frac{1}{4}$ x) $\frac{1}{2}$ y) $\frac{1}{4}$ z) $\frac{1}{2}$
6. a) 0 b) 4 c) 4 d) 4 e) 4 f) 4 g) 4 h) 4 i) 4 j) 4 k) 4 l) 4 m) 4 n) 4 o) 4 p) 4 q) 4 r) 4 s) 4 t) 4 u) 4 v) 4 w) 4 x) 4 y) 4 z) 4
7. a) 3 b) 4 c) 5 d) 6 e) 7 f) 8 g) 9 h) 10 i) 11 j) 12 k) 13 l) 14 m) 15 n) 16 o) 17 p) 18 q) 19 r) 20 s) 21 t) 22 u) 23 v) 24 w) 25 x) 26 y) 27 z) 28
8. a) 1 b) 2 c) 3 d) 4 e) 5 f) 6 g) 7 h) 8 i) 9 j) 10 k) 11 l) 12 m) 13 n) 14 o) 15 p) 16 q) 17 r) 18 s) 19 t) 20 u) 21 v) 22 w) 23 x) 24 y) 25 z) 26
9. a) 2 b) 3 c) 4 d) 5 e) 6 f) 7 g) 8 h) 9 i) 10 j) 11 k) 12 l) 13 m) 14 n) 15 o) 16 p) 17 q) 18 r) 19 s) 20 t) 21 u) 22 v) 23 w) 24 x) 25 y) 26 z) 27
10. a) 1 b) 2 c) 3 d) 4 e) 5 f) 6 g) 7 h) 8 i) 9 j) 10 k) 11 l) 12 m) 13 n) 14 o) 15 p) 16 q) 17 r) 18 s) 19 t) 20 u) 21 v) 22 w) 23 x) 24 y) 25 z) 26
11. a) 1 b) 2 c) 3 d) 4 e) 5 f) 6 g) 7 h) 8 i) 9 j) 10 k) 11 l) 12 m) 13 n) 14 o) 15 p) 16 q) 17 r) 18 s) 19 t) 20 u) 21 v) 22 w) 23 x) 24 y) 25 z) 26
12. The equation is true for all values of

4.7 Exponential Functions – Applications

Exponential Growth

$$y = a \cdot b^x$$

a is the initial amount.

b is the growth factor when $b > 1$.

If we rewrite b as $(1 + r)$, we can determine the GROWTH RATE, r. r is usually a percentage and written as a decimal.

$$y = a(1 + r)^x$$

eg. $y = 2(1.43)^x$

Growth factor _____

Growth rate _____

Initial value _____

Exponential Decay

$$y = a \cdot b^x$$

a is the initial amount.

b is the decay factor when $0 < b < 1$.

If we rewrite b as $(1 - r)$, we can determine the DECAY RATE, r. r is usually a percentage and written as a decimal.

$$y = a(1 - r)^x$$

eg. $y = 300(0.75)^x$

Decay factor _____

Decay rate _____

Initial value _____

1. A class of 26 students carried out an investigation to model population change. They decided that their data could be modeled by the equation $f(x) = 26(0.85)^x$, where f represents the number of people standing and x represents the number of trials.
 - a) Explain the equation (in other words, compare it to the $y = a \cdot b^x$ equation to state what each value represents).
 - b) Use the formula to calculate the number of students still standing after 10 trials.

2. Sally is training to run the Toronto Marathon. She starts running 5 km for the first week and increases the distance run by 10% every week.

a) What distance will she run during her 6th week of training?

b) Approximately, how many weeks will it take for Sally to run a half-marathon (21.1 km)?

Doubling Period

Bacterial and viral cultures are examples of substances that grow at a rate which is exponential in nature - they DOUBLE over a given period.

In general, these cultures grow according to the following exponential equation:

$$y = a(2)^{\frac{x}{d}}$$

y is _____

a is _____

x is _____

d is _____

Half-Life

The HALF-LIFE of a radioactive material (an isotope) is the time it takes for a sample to decay to half of the original amount.

In general, radioactive materials decays according to the following exponential equation:

$$y = a\left(\frac{1}{2}\right)^{\frac{x}{h}}$$

y is _____

a is _____

x is _____

h is _____

For example, suppose you have a petri dish containing 10 bacteria cells that double every hour,

after the 1st hr, there will be ____ cells

after the 2nd hr, there will be ____ cells

after the 3rd hr, there will be ____ cells

after the 4th hr, there will be ____ cells

For example, suppose you have a radioactive isotope that has a mass of 64 mg,

after one half-life, ____ mg is left

after two half-lives, ____ mg is left

after three half-lives, ____ mg is left

after four half-lives, ____ mg is left

Ex 3. One bacterium divide into two bacteria every 5 days. Initially, there are 15 bacteria. How many bacteria will there be in 30 days?

Ex 4. A certain radioactive material has a half-life of 35 days. If 100 g is present now, how many grams will be present in 1 year?

Ex 5. A bacterial culture starts with 3000 bacteria and grows to a population of 12 000 after 3 hours.
a) Find the doubling period.

b) Find an expression to represent the population after x hours.

c) Determine the number of bacteria after 8 hours.

Ex 6. A radioactive isotope, iodine-131, is used to determine whether a person has a thyroid deficiency. The iodine-131 is injected into the blood stream. A healthy thyroid gland absorbs all of the iodine. The half-life of iodine-131 is 8.2 days. After how long would 25% of the iodine-131 remain in the thyroid gland of a healthy person?

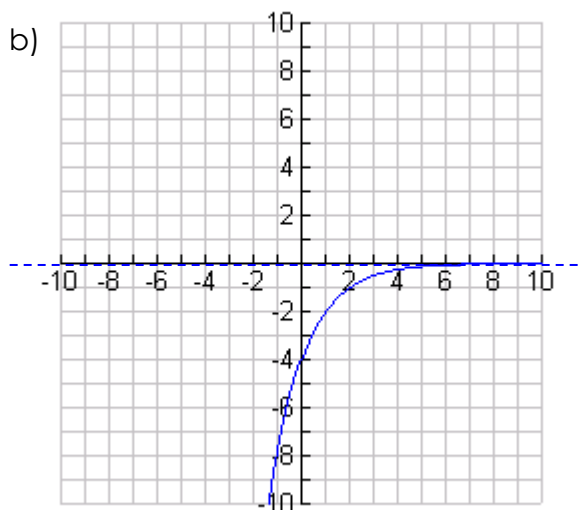
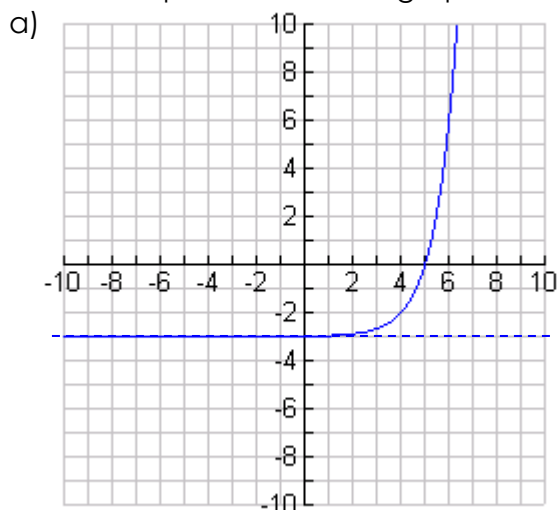
Word Problems Practice: Exponential Equations

Where necessary, round answers to 2 decimal places.

1. A bacterial culture doubles in size every 15 minutes. How long will it take for a culture of 20 bacteria to grow to a population of 163840? **[195 minutes]**
2. A colony of bacteria doubles in number every 20 minutes. The initial population of the colony is 20 bacteria. How many bacteria are there after 4 hours? **[81920]**
3. A sample of 700 cells in a medical research lab triples every 30 minutes. How long will it take for the population to reach 18900? **[90 minutes]**
4. A certain radioactive material has a half-life of 23 minutes. If 300 g are present now, approximately how many grams will be present after 2.5 hours? **[3.26 g]**
5. The half-life of ruthenium-106 is 1 year. If an original sample of ruthenium-106 has a mass of 128 mg, and there are 2 mg left, what is the elapsed time? **[6 years]**
6. An isotope has a half-life of 36 days. After a given period of time, $\frac{1}{256}^{\text{th}}$ of the original sample is remaining. How much time has passed? **[288 days]**
7. A bacterial culture starts with 5000 bacteria. After 45 minutes, there are 40000 bacteria. What is the doubling period? **[15 minutes]**
8. The world population is doubling about every 35 years. In 1980, the total population was about 4.5 billion. If the doubling period remains at 35, find the projected world population for the year 2120. **[72 billion]**
9. A viral culture doubles every $\frac{1}{4}$ of an hour. After $1\frac{1}{4}$ hours, there are 40000 cells.
 - a) What was the initial size of the culture? **[1250]**
 - b) What was the population after 3 hours? **[5120000]**
10. An isotope of sodium, Na^{24} , has a half-life of 15 hours.
 - a) Find the amount remaining from a 4-gram sample after 5 days. **[0.02 grams]**
 - b) How many hours will it take to decay to 2^{-4} grams? **[90 hours]**
11. The half-life of Palladium-100, Pd^{100} , is 4 days. After 16 days, a sample of Pd^{100} has been reduced to a mass of 0.75 grams.
 - a) What was the initial mass of the sample? **[12 grams]**
 - b) What is the mass after 2 days? **[8.49 grams]**

Review: Unit 4 – Exponential Functions

1. Find the equation from the graph.



2. An exponential function with a base of 4 has been compressed horizontally by a factor of $\frac{1}{3}$ and reflected in the x-axis. Its asymptote is the line $y = 3$. Its y-intercept is $(0, 2)$. Write an equation of the function and state its domain and range.

3. State the domain and range of $y = -3(2^{\frac{x}{2}+6}) - 1$.

4. Let $f(x) = 2^x$. For $g(x) = 4f(2x - 4) - 5$:

- state the equation of the asymptote and the y-intercept;
- state the domain and range.

5. Use the exponent rules to simplify each of the following. Express answers with positive exponents only.

a) $-a^7b^8c, a^2b^8c^6$

b) $\left(-\frac{1}{2}c^{-2}d^3\right)^{-4}$

c) $\frac{(5b^4)^3(25b^{-4})}{(5^{-1}b)^2}$

d) $\frac{x(x^{4a+1})}{x^{a+3}}$

e) $x^{3n+5}(x^{2n})$

f) $(4m^2n^{-3})\left(\frac{3m^{-4}n^{-2}}{2mn^2}\right)$

g) $\sqrt[4]{\frac{\sqrt{m} \times \sqrt[4]{m^5}}{\sqrt[3]{m^4}}}$

h) $\left[(c)^{2n-3m}\right](c^3)^m, (c^2)^n$

6. Evaluate each of the following.

a) $\frac{2^{-1}+4^{-2}}{2^{-2}+4^{-1}}$

b) $2^{-n}(2^n - 2^{1+n})$

c) $2\left(2^{3x} - \frac{1}{2^{-3x}}\right)$

7. Solve for x.

a) $\sqrt{\frac{9^{x+3}}{27^x}} = 81$

b) $121^{x-2} = 11\sqrt{11}$

c) $2^{x+1} \times 4^{x+1} \times 8^{x+1} = 128^x$

d) $\left(\frac{1}{16}\right)^{\frac{1}{4}} - \sqrt[3]{\frac{8}{27}} = \sqrt{x^2}$

e) $3^{x+1} + 3^x = 324$

8. Solve each of the following:

a) $4^{x+5} + 4^{x+4} = 160$

b) $-5(3^{2x+1}) + 4 = -401$

c) $\frac{64^x}{16^{x-2}} = 256$

d) $5^{x^2-2x} - 3(5^{x^2-2x}) = -250$

9. A population of yeast cells doubles in 2 hours. Assume an initial population of 120 cells.

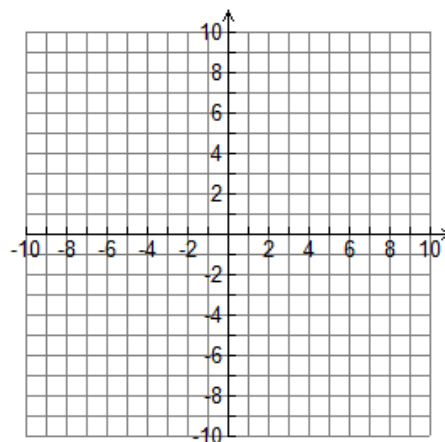
- Write an equation that can be used to determine the population of cells, M , at t hours.
- Use your equation to determine the population after 12 hours.
- Use your equation to determine the population after 210 min.
- Approximately how many hours would it take for the population to reach 498000?
- What are the domain and range for this situation?

10. A bone is tested and found to have 2 mg of C-14. The original amount of C-14 present in the bone was 16 mg. If the half-life of C-14 is 6000 years, how old is the bone?

11. Sodium-24, which has a half-life of 14.5 hours, is used to diagnose circulatory problems. How long would it take for the sample to decay to 25% of the original amount?

12. In 30 hours, a sample of plutonium decays to $\frac{1}{64}$ th of its original amount. What is the half-life of the substance?

13. Graph $f(x) = -2^{2x-2} + 1$ on the grid provided.



14. Find x and y if $\frac{3^{2x-y}}{27^x} = \frac{1}{27}$ and $\frac{5^{-x}}{125^{x+y}} = 25$.

Solutions: 1a) $f(x) = 3^{x-4} - 3$ b) $f(x) = -2^{-x+2}$ 2. $f(x) = -4^{3x} + 3$; $D = \{x \in \mathbb{R}\}$, $R = \{y \in \mathbb{R} \mid y < 3\}$

3. $D = \{x \in \mathbb{R}\}$, $R = \{y \in \mathbb{R} \mid y < -1\}$ 4a) $y = -5$; $y\text{-int} = -\frac{19}{4}$ b) $D = \{x \in \mathbb{R}\}$, $R = \{y \in \mathbb{R} \mid y > -5\}$ 5a) $-\frac{a^5}{c^5}$

b) $\frac{2^4 c^8}{d^{12}}$ c) $5^7 b^6$ d) x^{3a-1} e) x^{5n+5} f) $\frac{6}{m^3 n^7}$ g) $m^{\frac{5}{48}}$ h) 1 6a) $\frac{9}{8}$ b) -1

c) 0 7a) -2 b) $\frac{11}{4}$ c) 6 d) $-\frac{1}{6}$ e) 4 8a) $-\frac{3}{2}$ b) $\frac{3}{2}$ c) 0 d) 3 or -1 9a)

$M = 120(2)^{\frac{t}{2}}$ b) 7680 c) about 404 d) about 24 hours e) $D = \{t \in \mathbb{R} \mid t \geq 0\}$, $R = \{M \in \mathbb{I} \mid M \geq 120\}$ 10.

18000 years old

11. 29 hours 12. 5 hours

13

14. $(-11, 14)$

